

25. A black hole can be formed from a star which has become so massive it collapses under its own gravitational force.

The radius r of a black hole is calculated using the relationship

$$r = \frac{2GM}{c^2}$$

where: G is the Universal Constant of Gravitation

M is the mass of the star

c is the speed of light.

A star of mass 1.99×10^{31} kg collapses to form a black hole.

The radius of this black hole is

- A 7.37×10^3 m
- B 1.47×10^4 m
- C 2.95×10^4 m
- D 8.85×10^{12} m
- E 4.33×10^{15} m.